



Multi-Asset Signal Generation

HIERARCHICAL CLUSTER ANALYSIS

March 4, 2020

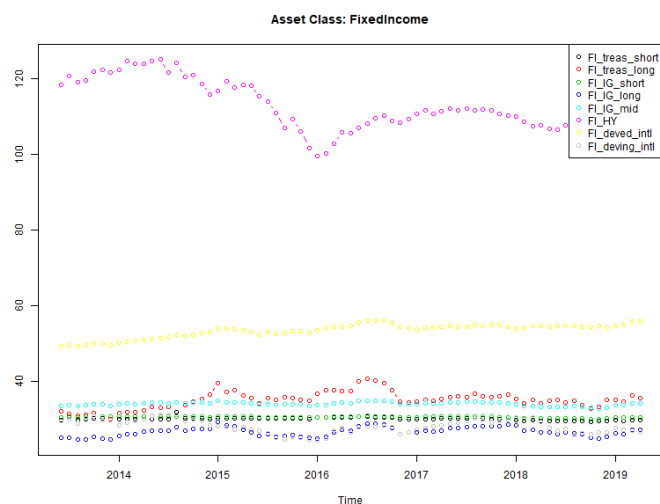
Principal Financial Group
Carnegie Mellon University

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1 Clusters Analysis of Asset Classes

1.1 Asset Class: *FixedIncome*



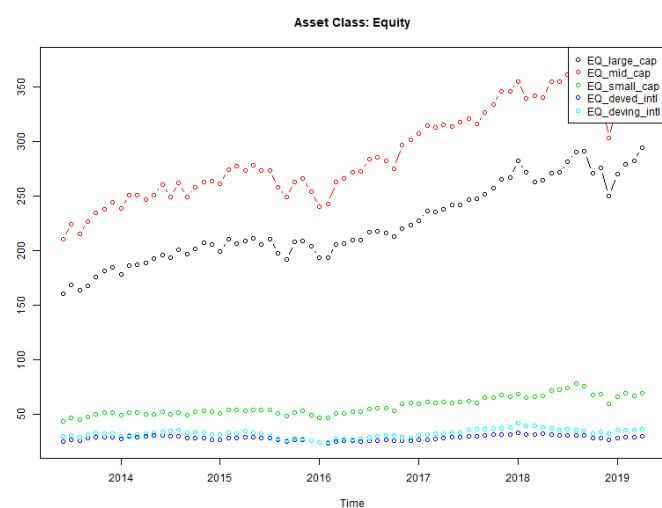
FixedIncome

ariance%	FI_deved_intl	FI_deving_intl	FI_HY	FI_IG_long	FI_IG_mid	FI_IG_short	FI_treas_long	FI_treas_short	Spearman	Kendall
64.9%	1	1	1	1	1	1	1	1	11%	7%
86.9%	2	1	1	2	2	2	2	1	27%	18%
86.9%	2	3	3	2	2	2	2	1	55%	38%
88%	2	4	4	2	3	3	2	1	59%	40%
88%	2	5	4	2	3	3	2	1	78%	63%
91.7%	5	6	4	2	3	3	2	1	83%	64%
97.6%	6	7	5	4	3	3	2	1	86%	69%
100%	7	8	6	4	5	3	2	1	100%	100%

Table 1: Cluster Analysis - FixedIncome

	1	2	3	4	5	6	7	8
FI_treas_short	1	1	1	1	1	1	1	1
FI_HY	1	1	3	4	4	4	5	6
FI_deving_intl	1	1	3	4	5	6	7	8
FI_treas_long	1	2	2	2	2	2	2	2
FI_IG_long	1	2	2	2	2	2	4	4
FI_deved_intl	1	2	2	2	2	5	6	7
FI_IG_short	1	2	2	3	3	3	3	3
FI_IG_mid	1	2	2	3	3	3	3	5
Variance% of PC1	64.9%	86.9%	86.9%	88%	88%	91.7%	97.6%	100%
Retained Correlation	11%	27%	55%	59%	78%	83%	86%	100%

1.2 Asset Class: *Equity*



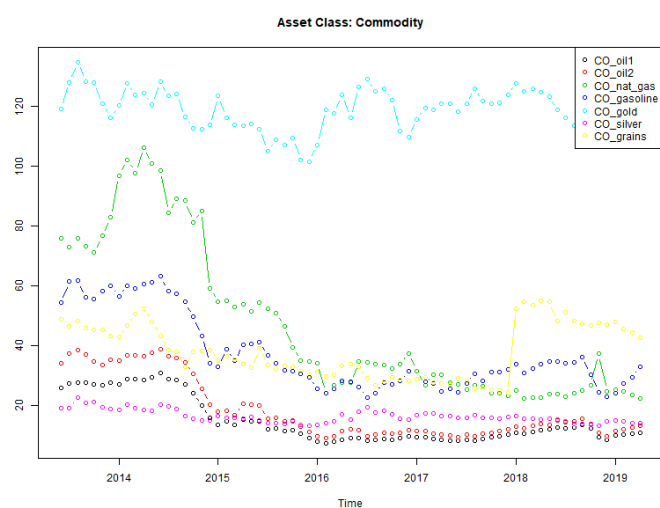
Equity

Variance%	EQ_deved_intl	EQ_deving_intl	EQ_large_cap	EQ_mid_cap	EQ_small_cap	Spearman	Kendall
95%	1	1	1	1	1	53%	38%
91.8%	2	2	1	1	1	88%	71%
95.8%	2	3	1	1	1	91%	74%
99.5%	3	4	1	2	2	93%	77%
100%	4	5	1	2	3	100%	100%

Table 2: Cluster Analysis - Equity

	1	2	3	4	5
EQ_large_cap	1	1	1	1	1
EQ_mid_cap	1	1	1	2	2
EQ_small_cap	1	1	1	2	3
EQ_deved_intl	1	2	2	3	4
EQ_deving_intl	1	2	3	4	5
Variance% of PC1	95%	91.8%	95.8%	99.5%	100%
Retained Correlation	53%	88%	91%	93%	100%

1.3 Asset Class: *Commodity*



Commodity

variance%	CO_gasoline	CO_gold	CO_grains	CO_nat_gas	CO_oil1	CO_oil2	CO_silver	Spearman	Kendall
38.9%	1	1	1	1	1	1	1	-1%	-1%
53.7%	1	2	1	1	1	1	2	28%	20%
65.2%	1	3	2	2	1	1	3	31%	21%
91.6%	1	3	4	2	1	1	3	84%	65%
97.9%	3	4	5	2	1	1	4	84%	65%
97.9%	3	4	6	2	1	1	5	97%	87%
100%	4	5	7	3	1	2	6	100%	100%

Table 3: Cluster Analysis - Commodity

	1	2	3	4	5	6	7
CO_oil1	1	1	1	1	1	1	1
CO_oil2	1	1	1	1	1	1	2
CO_gasoline	1	1	1	1	3	3	4
CO_nat_gas	1	1	2	2	2	2	3
CO_grains	1	1	2	4	5	6	7
CO_gold	1	2	3	3	4	4	5
CO_silver	1	2	3	3	4	5	6
Variance% of PC1	38.9%	53.7%	65.2%	91.6%	97.9%	97.9%	100%
Retained Correlation	-1%	28%	31%	84%	84%	97%	100%